const fs = require('fs'); const path = require('path'); const { execSync } = require('child_process'); async function generatePdf() { console.log('Generating CODAK Master Feature Workflow Dossier PDF...'); const mdPath = '/Users/sidd/Documents/codak_master_feature_workflow_dossier.md'; const mdContent = fs.readFileSync(mdPath, 'utf8'); // HTML Template with Executive Styling and Print CSS const htmlContent = `



CODAK

ACADEMIC GAMIFICATION & CODE ARENA ENGINE

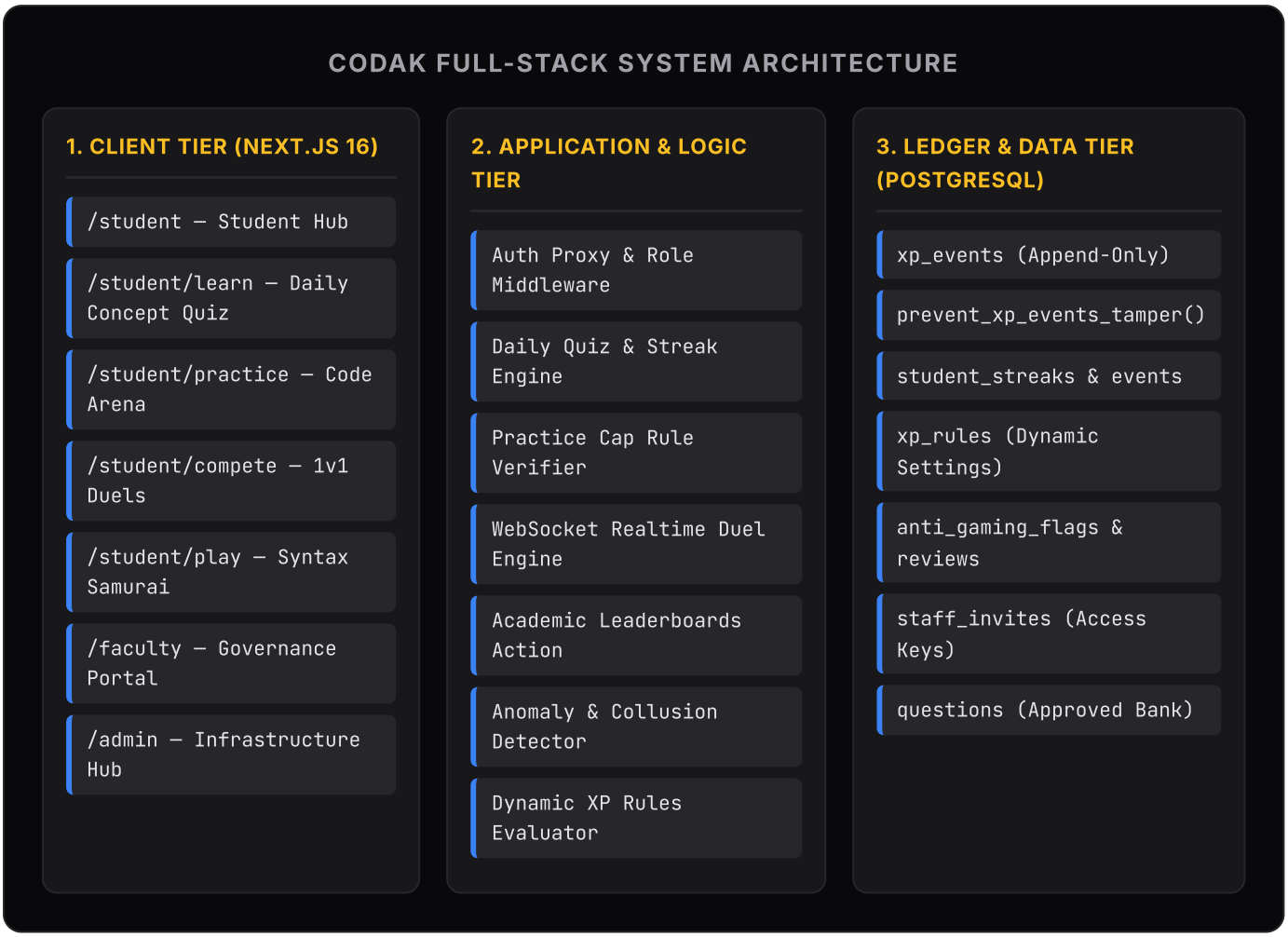
EXECUTIVE TECHNICAL DOCUMENTATION • V1.0 PRODUCTION

Master Feature & Architectural Workflow Dossier

A comprehensive technical blueprint of the student gamification lifecycle, faculty academic governance, cryptographic XP ledger integrity, dedicated campus leaderboards, and anti-gaming detection engines.

Core Framework	Database & Security	Production Status	Release Date
Next.js 16 (App Router & Turbo)	Supabase PostgreSQL with RLS	39 Static & Dynamic Routes Verified	October 20, 2026

Platform Vision & Philosophy: CODAK bridges student intrinsic motivation with faculty academic governance. Every XP point is backed by an append-only, tamper-proof PostgreSQL ledger, while autonomous statistical detection algorithms protect academic integrity across single-campus and multi-institution networks.



1. Student Experience & Gamification Lifecycle

1.1 Authentication & Campus Onboarding

- **Multi-Provider Auth:** Supports high-security Email/Password authentication as well as Google OAuth One-Tap sign-in.
- **Campus Selection:** Unaffiliated students are guided to `/select-campus` to enroll under an institution (e.g., *Rathinam Institute of Technology*) or custom institution. Captures Department, Roll Number, and Year of Study (1–4).

1.2 Student Dashboard & Activity Heatmap (`/student`)

- **Level Velocity Equation:** Non-linear equation ensuring progressive effort: `Level(XP) = floor(1 + sqrt(XP / 25))`.
- **Rank Tiers:** Celebrates student milestones across 6 badges: *Apprentice* (Lv. 1–3), *Novice* (Lv. 4–7), *Adept* (Lv. 8–12), *Specialist* (Lv. 13–18), *Master* (Lv. 19–25), and *Grandmaster* (Lv. 26+).
- **Live Streak Pill with Active Decay:** Tracks consecutive calendar days in Asia/Kolkata (IST). If a student misses days beyond their monthly freeze allowance, the UI immediately decays the active streak to 0.

- **31-Day Activity Heatmap:** GitHub-style visual calendar mapping valid daily XP earnings in local IST time.

1.3 Daily Concept Quiz (`/student/learn`)

- **Curated Daily Challenge:** 5 curriculum questions drawn from faculty-approved questions.
- **Strict Alphabetical Order:** Options A, B, C, and D are maintained in sequential alphabetical order without disruptive shuffling.
- **Information Hiding:** Correct options and detailed explanations are stripped server-side until the attempt is submitted.
- **Scoring & 7-Day Milestone:** Base 15 XP + 5 XP per correct answer. Consecutive completion on Day 7 triggers a 50 XP milestone bonus (`STREAK_7_BONUS`). Up to 2 freeze days per month automatically protect broken streaks.

1.4 Practice Arena (`/student/practice`)

- **Categorized Problem Bank:** Topics in Data Structures, Algorithms, AI/ML, and Core CS with Easy (10 XP), Medium (20 XP), and Hard (40 XP) problem sets.
- **External Links:** Direct launches to LeetCode, HackerRank, and GeeksforGeeks.
- **Daily Practice Cap:** Hard 40 XP daily limit enforced inside the database stored procedure to prevent scripting.

1.5 1v1 Real-Time Code Duels & Battle Arena (`/student/compete`)

- **WebSocket Matchmaking:** Live pairing of students in competitive timed battles.
- **Dual Health Bars & Real-Time Telemetry:** Opponent answer speed and life bar updates are streamed in real time.
- **Battle Payouts:** `DUEL_WIN` (25 XP), `DUEL_PARTICIPATION` (10 XP), `BATTLE_WIN` (35 XP), and `BATTLE_PARTICIPATION` (15 XP).

1.6 Retro Cyber Arcade & Mini-Games (`/student/play`)

- **Codemons Arcade:** Retro monster creature showcase and cyber lounge.
- **Syntax Samurai Dojo (`/student/play/syntax-samurai`):** Katana cursor blade with dynamic slash trail, pixel-crumble particle effects, syntax bug slicing, and Code Review Victory screen explaining bugs pedagogically.

2. Faculty Academic Oversight & Governance

2.1 Academic Analytics Dashboard (`/faculty`)

- **Cohort Health Metrics:** 7-day active rate, daily quiz completion rate, average cohort streak, and department velocity.
- **Topic Weakness Diagnostics:** Automatically flags topics with <50% success rates across student attempts.
- **Quiet Student Detection:** Identifies disengaged students inactive for >3 days for early academic support.
- **Live Student Directory:** Searchable student roster complete with lifetime XP, rank badges, and active streaks.

2.2 Question Curation & "Approve All" Batch System (`/faculty/questions`)

- **Question Submission:** Submit new multiple-choice questions with topics, difficulty, explanations, and four options.
- **"Approve All" Action:** Modal confirming batch approval of all pending question submissions in a single transaction.

2.3 Manual XP Awarding (`/faculty/award-xp`)

- **Academic Activity Allocation:** Class Attendance (10 XP), On-Time Assignment (20 XP), Peer Tutoring (15 XP), Technical Club (25 XP), and Hackathon Participation (40 XP).
- **Audit Justification:** Requires written academic reason (`award_reason`) permanently attached to the ledger record.

2.4 HackerRank Programming Contests (`/faculty/competitions`)

- **Contest Setup:** Schedule campus coding contests with time windows and HackerRank URLs.
- **Bonus Allocation:** Awards 25 XP (`HACKERRANK_COMPETITION`) to verified contest participants.

2.5 Academic Integrity & Anti-Gaming Hub (`/faculty/flags`)

- **Autonomous Detection Telemetry:** Detects XP spikes (>3x baseline), answer-sharing collusion, and duel win-trading.
- **Zero Auto-Void Rule:** Telemetry never cancels points automatically. Points can only be voided by human faculty with mandatory written justification.

3. Dedicated Academic Leaderboards & Comparative Analytics

Accessible directly via `/faculty/leaderboard` and `/admin/leaderboard` (no longer redirecting to student views).

3.1 Inter-Campus Comparison (Standings & Averages)

- **Fair Metric Ranking:** Campuses are ranked by **Average XP per Student** to eliminate size bias between large and small institutions.
- **System Benchmark Comparison:** Highlights whether each campus is performing **Above Global Average** (+XX% in emerald) or **Below Global Average** (-XX% in amber).
- **Institution Metrics:** Total registered, active players, total XP, and top scorer peak.

3.2 Intra-Campus Rankings & Department Averages

- **Campus & Batch Selectors:** Dropdowns to switch across campuses, departments (CSE, AIML, ECE, Mechanical), and study years (1–4).
- **Department Averages Strip:** Visual benchmark cards showing each department's average XP and its delta from the campus average.
- **Student Roster:** Real student identities, roll numbers, medals (#1 Gold, #2 Silver, #3 Bronze), streaks, levels, and delta badges (Above/Below Avg).

3.3 Higher vs Lower Average Matrix

- **Higher Average Quadrant:** Isolates top cohorts operating above baseline standards.
- **Lower Average / Growth Quadrant:** Highlights cohorts trailing baseline to guide faculty in deploying targeted practice links or challenge contests.

4. Admin Infrastructure & Ledger Audit

4.1 Admin Hub & Staff Directory (`/admin`)

- **Comprehensive Student Directory:** Full directory of all registered students with real-time search.
- **Faculty Directory with Access Key Attribution:** Lists all faculty members with department, designation, and **which Staff Invite Access Key was redeemed** to establish their credentials.

4.2 Cryptographic Staff Invites (`/admin/invites` & `/redeem-invite`)

- **Access Key Generator:** Generates cryptographic, single-use invite tokens for Faculty and Administrator onboarding.
- **Redemption Tracking:** Live status tracking showing `Active`, `Expired`, or `Redeemed by: Name (email)` with timestamp.

4.3 Dynamic XP Rules Engine (`/admin/rules`)

- **Hot Configuration:** Live editing of base XP, extra XP, daily occurrence caps, and active/disabled toggles across all 15+ activities without re-deploying code.

4.4 Ledger Audit & Anti-Tamper Verification (`/admin/audit`)

- **Immutable Append-Only Ledger:** Inspect the raw `xp_events` table.
- **Trigger Defense Check:** Verifies that database trigger `prevent_xp_events_tamper()` is active, blocking direct SQL UPDATE or DELETE queries.
- **Void Audit Trail:** Complete log of faculty-voided points displaying actor, timestamp, void reason, and reference ID.

5. Anti-Gaming Architecture & Security Guarantees

SECURITY CHALLENGE	ATTACK VECTOR	CODAK DEFENSE MECHANISM
Replay / Double Payout	Double-tapping submit button or replaying network request	Database UNIQUE constraint `unique_student_action_ref`. Replays return `status: duplicate` and 0 XP.
Direct Client Tampering	Malicious student modifying points via Supabase client	RLS blocks client mutations. Trigger `prevent_xp_events_tamper()` throws fatal exception on direct UPDATE/DELETE.
Streak Farming	Submitting multiple quizzes on the same calendar day	Engine identifies same-day completion; streak increments strictly once per Asia/Kolkata (IST) calendar day.
Answer Collusion	Student pairs submitting identical answers simultaneously	Statistical correlation engine analyzes wrong-choice overlap; flags pairs with >80% shared anomaly score for faculty audit.
Practice Farming	Submitting infinite fake practice claims	Automated verification, duplicate proof checking, and hard 40 XP daily practice ceiling inside database stored procedure.
Duel Win-Trading	Two collusion accounts repeatedly challenging each other in 1v1	Duel frequency detector flags accounts with excessive mutual matches in a 7-day rolling window.

6. Complete Route Catalog (39 Routes)

ROUTE PATH	ACCESS ROLE	DESCRIPTION & PRIMARY FUNCTION
`/`	Public	Hero landing page showcasing features, live stats, and student login
`/login`	Public	Role-aware modern sign-in (Email/Password & Google OAuth)
`/select-campus`	Student	Onboarding screen for new students to choose institution & roll number
`/redeem-invite`	Staff	Access key redemption portal for faculty and admin onboarding
`/student`	Student	Main student hub: XP level, rank, streak calendar, fast action cards
`/student/learn`	Student	Daily Concept Quiz, 5 questions, streak progression, milestone bonus
`/student/practice`	Student	Practice problem directory, difficulty tags, claim submission
`/student/compete`	Student	1v1 code duels matchmaking lobby, league divisions, rankings
`/student/compete/battles/[id]`	Student	Live real-time WebSocket 1v1 battle arena
`/student/play`	Student	Retro cyber arcade hub featuring Codemons & Syntax Samurai
`/student/play/syntax-samurai`	Student	Katana slice mini-game for real-time syntax bug correction
`/student/quests`	Student	Weekly quest challenges with milestone XP payouts
`/student/profile`	Student	Edit student profile, avatar customization, anonymous mode toggle
`/faculty`	Faculty	Faculty dashboard: engagement rates, weak topics, student roster
`/faculty/questions`	Faculty	Question bank curation, approval queue, and **Approve All** button
`/faculty/award-xp`	Faculty	Manual XP allocation with mandatory written academic justifications
`/faculty/competitions`	Faculty	HackerRank programming contests management and payouts
`/faculty/flags`	Faculty	Anti-Gaming & Integrity Flags hub for anomaly reviews and voids
`/faculty/leaderboard`	Faculty	Dedicated inter-campus, intra-campus, and average comparison matrix
`/admin`	Admin	Admin hub: full student roster, faculty directory with key attribution
`/admin/invites`	Admin	Staff invite generation, cryptographic access key management
`/admin/rules`	Admin	Dynamic XP rules engine: live edit base XP, extra XP, and limits

ROUTE PATH	ACCESS ROLE	DESCRIPTION & PRIMARY FUNCTION
`/admin/audit`	Admin	Immutable ledger auditor, void history, anti-tamper verification
`/admin/flags`	Admin	Centralized security and statistical anomaly flags hub
`/admin/leaderboard`	Admin	Dedicated cross-campus comparative analytics and averages
`/api/cron/detect-spikes`	System/Cron	Autonomous cron detecting daily XP spikes >3x baseline
`/api/cron/detect-collusion`	System/Cron	Autonomous cron analyzing answer-sharing and win-trading
`/api/cron/weekly-leagues`	System/Cron	Autonomous cron calculating weekly league promotions/relegations
`/api/auth/signout`	Authenticated	Server-side secure session termination and cookie cleanup

7. Verified Review & Evaluation Accounts

All review accounts share the standard password: `Password123!`

```
Faculty Review Account: faculty@demo.edu / Password123!
Admin Review Account:   admin@demo.edu   / Password123!
Student Review Account: student@demo.edu / Password123!
```